

VISHAL PATEL

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Skills Summary

Applied Mathematics Graduate with a focus on Statistical Learning and Data Science from the University of Pittsburgh, possessing exceptional quantitative and analytical problemsolving skills. Experienced in Data Engineering and Python development with Big Data technologies, particularly in AWS, and proficient in SQL programming. Seeking a Data Engineer role in a data-driven company to develop and optimize data pipelines and data warehouse architecture.

Education

BS in Applied Mathematics from University of Pittsburgh

Experience

Data Engineer at Penn Entertainment, Wyomissing, PA March 2021 — Present

- Participating in daily stand-ups and meetings involving requirements analysis and design. Working with Git, and Jira in a team environment which follows Agile Methodologies.
 - Developed and integrated monitoring for the cloud platforms and data pipelines to ensure the highest levels of data quality and availability.
 - Partnering with the Business Analytics Team, and internal technology teams to collect and document solution requirements, data access, process automation, and performance objectives.
 - Created data pipelines that utilize AWS Kinesis to ingest data in real-time and process using AWS Lambda.
 - Worked on Python Scientific packages like NumPy and Pandas to handle data sets effectively and monitored trading systems and daily data processing.
 - Created AWS Lambda functions that process real-time data and write to securely managed RDS instances in AWS, with the use of ElastiCache to minimize latency.
 - Manage and maintain databases using technologies such as PostgreSQL and MySQL.
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- Helped set up processes to maintain and store data that is non-relational using software such as DynamoDB.

Python Developer at Horizon Media, New York City, NY August 2019 — March 2021

- Participating in daily stand-ups and meetings involving requirements analysis and design. Working with Git, and Jira in a team environment which follows Agile Methodologies.
- Implemented solutions for ingesting data from various sources and processing the data utilizing Big Data technologies such as Hadoop, Map Reduce Frameworks, and Hive.
- Wrote Lambda functions in python for AWS's Lambda which invokes python scripts to perform various transformations and analytics on large data sets in Amazon EMR for processing Big Data across a Hadoop Cluster of virtual servers on Amazon Elastic Compute Cloud (EC2) and Amazon Simple Storage Service (S3).
- Wrote and executed several complex SQL queries in AWS glue for ETL operations in Spark data frame using sparkSQL.
- Managed code versioning with GitHub, Bit Bucket and deployment to staging and production servers.
- Installed various applications on AWS EC2 instances and configured the storage on S3 buckets and used Spark-context to analyze dataset and explore data using lambda function.
- Created an automated program using Python and various AWS tools such as SNS and Athena, that sends a daily alert of daily updates to all databases in Amazon Athena managed by the company.
- Managing EC2 instances, debugging python applications on instances, setting up daily batch jobs, scaling up/down specs of EC2 instances, attaching EBS volumes (CentOS7, Python 3.6).
- Architecting, tuning, and writing PySpark code to read in several 100 MBs of JSON, perform ETL operations, and generate parquet files in S3. Converting python code to PySpark code reduced runtime by several hours (EMR, Python 3.6).
- Helped implement Airflow to automate data pipelines and workflows.